

Loïc Dubois

Born in 1999
French citizen
ldubois@nd.edu
loic-dubois.github.io

Employment

Oct. 2025- University of Notre Dame, USA. Postdoc in the group of Erin Chambers.
2022-2025 Université Gustave Eiffel, France. Research and teaching grant for PhD preparation.
2018-2022 École Normale Supérieure de Lyon, France. Training salary.

Education and diploma

2025 PhD thesis: *Algorithms for topological and metric surfaces*. Université Gustave Eiffel, France.
Supervised by Éric Colin de Verdière and Vincent Despré.
2021-2022 Internships and diploma. École Normale Supérieure de Lyon.
2019-2021 Master: Theoretical Computer Science. École Normale Supérieure de Lyon, France.
2018-2019 Bachelor: Theoretical Computer Science. École Normale Supérieure de Lyon, France.
Bachelor: Mathematics for Engineering. Université Claude Bernard, France.

Awards

2022 Best Student Paper Award, Canadian Conference on Computational Geometry (CCCG).

Research internships

2021-2022 Université Gustave Eiffel, France. Supervised by Éric Colin de Verdière and Vincent Despré.
Subject: untangling graphs on surfaces. *Leading to [A1, A2, A3]*.
Technische Universität Berlin, Germany. Supervised by Stefan Felsner. Subject: extremal
trees for the Wiener index.
2021 INRIA Nancy, France. Supervised by Vincent Despré and Monique Teillaud. Subject: Delau-
nay flips algorithms on hyperbolic surfaces. *Leading to [A4, A6][B1]*.
2020 Universitat Politècnica de Catalunya, Spain (remote due to covid). Supervised by Guillem
Perarnau. Subject: centered colorings of graphs. *Leading to [A5]*.
2019 ONERA Toulouse, France. Supervised by Pierre-Loïc Garoche. Subject: approximation by
semi-algebraic sets.

Implementation

- [B1] Vincent Despré, Loïc Dubois, Marc Pouget, and Monique Teillaud. 2D triangulations on hyperbolic surfaces. In *CGAL User and Reference Manual*. CGAL Editorial Board, 6.1 edition, 2025. URL: <https://doc.cgal.org/6.1/Manual/packages.html#PkgHyperbolicSurfaceTriangulation2>.

Publications

- [A1] Éric Colin de Verdière, Vincent Despre, and Loïc Dubois. Untangling graphs on surfaces. *Journal of the ACM*, 2026.
- [A2] Éric Colin de Verdière, Vincent Despré, and Loïc Dubois. A discrete analog of tutte’s barycentric embeddings on surfaces. *TheoretiCS*, 5, 2026.
- [A3] Loïc Dubois. Making multicurves cross minimally on surfaces. In *32nd Annual European Symposium on Algorithms (ESA 2024)*, volume 308, pages 50:1–50:15, 2024. doi:10.4230/LIPIcs.ESA.2024.50.
- [A4] Loïc Dubois. Computing the Intrinsic Delaunay Triangulation of a Closed Polyhedral Surface. In Hee-Kap Ahn, Michael Hoffmann, and Amir Nayyeri, editors, *42nd International Symposium on Computational Geometry (SoCG 2026)*, volume 367 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 40:1–40:15, Dagstuhl, Germany, 2026. Schloss Dagstuhl – Leibniz-Zentrum für Informatik. URL: <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.SocG.2026.40>, doi:10.4230/LIPIcs.SocG.2026.40.
- [A5] Loïc Dubois, Gwenaél Joret, Guillem Perarnau, Marcin Pilipczuk, and François Pitois. Two lower bounds for p -centered colorings. *Discrete Mathematics & Theoretical Computer Science*, 22(Graph Theory), 2020. doi:10.23638/DMTCS-22-4-9.
- [A6] Loïc Dubois. A bound for Delaunay flip algorithms on flat tori. *Computing in Geometry and Topology*, 2(2):6:1–6:13, 2023. doi:10.57717/cgt.v2i2.27.

External reviewing

Symposium on Discrete Algorithms (SODA) – Symposium on Computational Geometry (SoCG) – European Symposium on Algorithms (ESA) – Graph Drawing (GD) – Symposium on Theoretical Aspects of Computer Science (STACS) – Discrete and Computational Geometry (DCG) – Journal of Computational Geometry (JoCG) – Computing in Geometry and Topology (CGT)

Teaching

Assembler (NASM) – OpenGL – SQL – Algorithms and programming in Python – Maths for computer scientists

Languages

Python – C++ – L^AT_EX – SQL – NASM